

ACC Two-Channel Torque Assembly

Physical Conditions — Smoothstep Gate Inputs

Proximity $|\Delta x|$ · Quietness $\text{EMA}(|v|)$ · Stability $\theta, \dot{\theta}$ · Contact Fz^L, Fz^R · Terrain $h^L_{\text{gnd}}, h^R_{\text{gnd}}$

Wheel Torque τ_w

τ_{balance}
Sagittal Balance Core
(pitch PD + vel. damp + pos-P
+ lateral + yaw)
always active

+

$g_{\text{anchor}} \cdot \tau_{\text{anchor}}$
Proximity-Gated Anchor
(integral + damping boost)
gated by $g_{\text{prox}}, g_{\text{env}}$

+

$g_{\text{flight}} \cdot \tau_{\text{flight}}$
Contact-Loss Recovery
(reaction-wheel attitude PD)
gated by g_{flight}



$$\tau_w = \tau_{\text{bal}} + g_{\text{anc}} \cdot \tau_{\text{anc}} + g_{\text{flt}} \cdot \tau_{\text{flt}}$$

Leg-Joint Torque τ_q

$\tau_{\text{posture}}(h_{\text{cmd}})$
Posture & Yaw Stability
(Jacobian PD + hip-yaw
divergence + homing)
always active, height-scheduled

+

$g_{\text{terrain}} \cdot \Delta \tau_{\text{posture}}$
Per-Leg Ground Adaptation
(split height commands
+ leveling integrator)
gated by g_{terrain}



$$\tau_q = \tau_{\text{post}} + g_{\text{terr}} \cdot \Delta \tau_{\text{post}}$$

Both channels assembled independently → final 10-DoF torque command